

## WEEK 2 PREREQUISITE ASSIGNMENT

# STEP 1: STOCK MARKET FUNDAMENTALS

A beginner-friendly overview with a Bluestock project case example

Purpose: understand basic stock-market terms, read the three main financial statements, and use simple ratios. This handout is for learning and is not investment advice.

## PART A - ELEVEN STOCK-MARKET CONCEPTS

### 1. Stock Market

A stock market is a marketplace where shares and other securities of listed companies are bought and sold. A share represents a small ownership interest in a company. Returns may come from a change in share price and from dividends.

### 2. NSE and BSE

NSE is the National Stock Exchange of India and BSE is the Bombay Stock Exchange. Both are major Indian stock exchanges. Nifty 50 is associated with NSE; Sensex is associated with BSE.

### 3. Nifty and Sensex

Nifty 50 tracks 50 major NSE-listed companies. Sensex tracks 30 major BSE-listed companies. They are benchmark indices used to observe broad market movement.

### 4. IPO and SME IPO

An IPO (Initial Public Offering) is when a private company first offers shares to the public. An SME IPO is an IPO route designed for eligible small and medium enterprises.

### 5. Market Capitalization

Market capitalization is the market value of all outstanding shares. Formula: Current share price x outstanding shares. Companies are often described as large-cap, mid-cap or small-cap.

## **PART A - ELEVEN STOCK-MARKET CONCEPTS (CONTINUED)**

### **6. P/E Ratio**

Price-to-Earnings ratio compares market price per share with earnings per share (EPS).

Formula:  $P/E = \text{market price per share} / \text{EPS}$ . It is most useful when compared with similar companies or the company's own history.

### **7. P/B Ratio**

Price-to-Book ratio compares market price per share with book value per share. Formula:  $P/B = \text{market price per share} / \text{book value per share}$ .

### **8. EPS**

Earnings Per Share shows profit attributable to each outstanding share. Basic formula:  $\text{profit attributable to equity shareholders} / \text{weighted average number of shares}$ .

### **9. Dividend**

A dividend is a distribution of part of a company's profits to shareholders. Example: 100 shares x INR 10 dividend per share = INR 1,000. A profitable company may choose to retain rather than distribute profit.

### **10. Bonus Shares and Stock Split**

Bonus shares are additional shares issued to existing shareholders without payment, subject to the company's terms. A stock split increases the number of shares and proportionately reduces the per-share price. Neither action automatically creates wealth.

### **11. Trading Volume**

Trading volume is the number of shares traded in a period. It helps show activity and liquidity. High volume may accompany strong buying or selling interest, but volume alone does not determine whether a stock is good or bad.

## **QUICK FORMULA REFERENCE**

Market Cap = Share Price x Outstanding Shares

$P/E = \text{Market Price per Share} / \text{EPS}$

$P/B = \text{Market Price per Share} / \text{Book Value per Share}$

$\text{EPS} = \text{Profit attributable to equity shareholders} / \text{Weighted average shares}$

# COMPANY FINANCIAL STATEMENTS

The three statements answer different questions about a company.

## 1. Balance Sheet

A balance sheet shows a company's financial position at a particular date. It lists what the company owns (assets), what it owes (liabilities), and the owners' interest (equity).  $\text{Assets} = \text{Liabilities} + \text{Equity}$ .

## 2. Profit and Loss Statement

The Profit and Loss statement shows performance over a period. It normally follows: Revenue - > Expenses -> Profit Before Tax -> Tax -> Profit for the Year.

## 3. Cash Flow Statement

A cash flow statement explains how cash changed during a period. It covers operating activities, investing activities and financing activities. Profit and cash are related but are not always the same.

# BLUESTOCK MUTUAL FUND ANALYTICS - CASE EXAMPLE

Bluestock is used here as a project case: it analyses mutual-fund data rather than publishing financial statements for a listed company.

## How the project uses market data

Fund master data identifies schemes, AMFI codes, fund houses and categories. NAV history supports return analysis. AUM and SIP data show managed assets and recurring investor participation. Investor transactions support cohort and SIP-continuity analysis. Holdings and benchmarks support sector-concentration and performance context.

## **BLUESTOCK - ANALYSIS EXAMPLES**

The following examples use project data categories, not invented company figures.

### **Fund performance**

Using NAV history, the project can calculate periodic returns, volatility and the Sharpe ratio. A higher Sharpe ratio generally indicates stronger return relative to the volatility taken, but it should be compared over the same period and category.

### **Downside risk**

Historical Value at Risk (VaR) and Conditional VaR (CVaR) can be calculated from daily fund returns. These measures help describe historical downside outcomes; they do not guarantee future losses.

### **Investor behaviour**

By grouping investors by first transaction year, Bluestock can compare cohorts. SIP continuity analysis can flag eligible investors whose average gap between SIP transactions is greater than the project threshold.

### **Fund recommendation**

The project's simple recommender matches a user's risk appetite (Low, Moderate or High) to a risk grade and lists the top funds by Sharpe ratio within that group.

### **Sector concentration**

Using portfolio-holdings weights, the Herfindahl-Hirschman Index (HHI) can show whether an equity fund is diversified across sectors or concentrated in fewer sectors.

## **Basic financial ratios - formula reference**

Net Profit Margin =  $\text{Net Profit} / \text{Revenue} \times 100$

Current Ratio =  $\text{Current Assets} / \text{Current Liabilities}$

Return on Equity =  $\text{Net Profit} / \text{Average Equity} \times 100$

These company-level ratios require an actual company's financial statements. They are included for prerequisite learning but are not calculated from the Bluestock mutual-fund datasets.

## **KEY FINDINGS AND CONCLUSION**

### **Market-data understanding**

The project brings together NAV, fund, AUM, SIP, investor, holdings and benchmark data so that related financial-market questions can be examined in one workflow.

### **Performance and risk**

NAV history supports return and risk measures such as rolling Sharpe ratio, historical VaR and CVaR.

### **Investor insight**

Investor-transaction and SIP data support cohort analysis and continuity monitoring.

### **Portfolio insight**

Portfolio holdings support sector-concentration analysis; benchmark data provides context for assessing fund performance.

### **Overall conclusion**

Bluestock Mutual Fund Analytics applies basic stock-market knowledge to a practical data-analysis project. It demonstrates how financial data can be cleaned, analysed and presented without making unsupported company-level claims or investment recommendations.

## **DATA-SCOPE NOTE**

This document uses Bluestock as a project example. Its examples are based on the project dataset categories: fund master, NAV history, AUM, SIP inflows, category inflows, folio counts, scheme performance, investor transactions, portfolio holdings and benchmark indices. No company financial figures are claimed or calculated for Bluestock.

## **SUBMISSION CHECKLIST**

[x] 11 stock-market concepts

[x] Balance Sheet, Profit and Loss Statement, and Cash Flow Statement

[x] Basic financial ratios

[x] Bluestock project example, analysis and conclusion